What Bars Cannot See: Order Flow, Its OHLCV Proxies, and Short-Horizon Bitcoin Returns

Seyed Abolfazl Hosseini, Ferdowsi University of Mashhad

Research summary, November 2026 Paper and code: github.com/a0x0h/what-bars-cannot-see

Question. Bulk volume classification (BVC), Kyle's lambda and high-low spread estimators are used to study order flow and liquidity when trade-level data are unavailable, and increasingly as features in return forecasts. How much of the true quantity do they capture, and does the true quantity or its proxy predict short-horizon returns out of sample and after costs?

Data and identification. Binance spot BTC/USDT and ETH/USDT, January 2022 to October 2026, hourly and one-minute bars (about 42,000 hours and 2.5 million minutes per asset). The exchange reports the taker-buy volume of every bar, so the true signed order flow is observed; on one month of trade-level data the bar-level imbalance matches trade-level aggressor flow with correlation 1.0000. A zero-fee promotion on BTC/USDT only (July 2022 to March 2023) gives a natural comparison with ETH/USDT.

Methods. (1) Proxy validation against observed flow and one-minute realised volatility, and a nonparametric *information ceiling*: the R^2 of the best possible function of the bar's standardised return, an upper bound for every return-based classifier including BVC and the tick rule. (2) Return response to one-SD flow shocks over 1–24 hours, separating the part of flow visible in the bar's price from the part that is not (Newey-West inference). (3) Monthly walk-forward over 53 test months with purged labels and all preprocessing and tuning inside each training window; elastic net and gradient boosting on nested feature sets; Clark-West, Diebold-Mariano and block-bootstrap inference; overlapping-sleeve trading rules with costs and the deflated Sharpe ratio. Automated tests verify that no feature uses future data.

Findings.

1. *Volatility is measured well, order flow is not.* Garman-Klass volatility from hourly bars has correlation 0.99 with realised volatility. BVC imbalance has correlation 0.49 with observed imbalance, and no function of the bar return explains more than 0.25 of its variance even in sample (0.34 on one-minute bars): most aggressor flow leaves no trace in the bar's price, BVC is already near this bound, and its scale drifts over time. Kyle's lambda from BVC has correlation 0.26 with lambda from true flow. The Corwin-Schultz spread has a median of 15.8 bps on a market whose tick is 0.002 bps and moves with volatility (correlation 0.87). Results are nearly identical for ETH.
2. *In BTC, the information in flow is in the part bars cannot see.* A one-SD taker-imbalance shock is followed by a price decline of 1.3 bps over two hours ($t = -4.0$), robust to lagged returns, volume and volatility and stable across fee regimes and sample halves. Holding the bar return fixed leaves it unchanged. Predictive content rises with measurement accuracy: none for hourly BVC, part for BVC on one-minute bars, all for observed flow, which subsumes the proxy. In ETH the effect is weaker and not significant.
3. *Detectable out of sample only when measured, and too small to trade.* Used alone in a monthly walk-forward, observed imbalance forecasts BTC returns out of sample ($R^2_{OOS} = 0.021\%$, Clark-West $p = 0.0001$ at two hours, significant in both halves); hourly BVC does not. Traded on its own, the signal breaks even at about 1.2 bp per unit of turnover. No multivariate model using microstructure proxies, observed flow or realised measures beats a zero forecast; cross-validation selects the empty model in most months. The best of 16 trading rules has a gross Sharpe ratio of 1.24 but a deflated Sharpe ratio of 0.70, and no rule breaks even at the standard fee.
4. *Fees change the data.* Binance's zero-fee promotion on BTC/USDT multiplied hourly trade counts fivefold and cut the dispersion of aggressor imbalance from 0.140 to 0.030; ETH/USDT, not included, was unaffected.

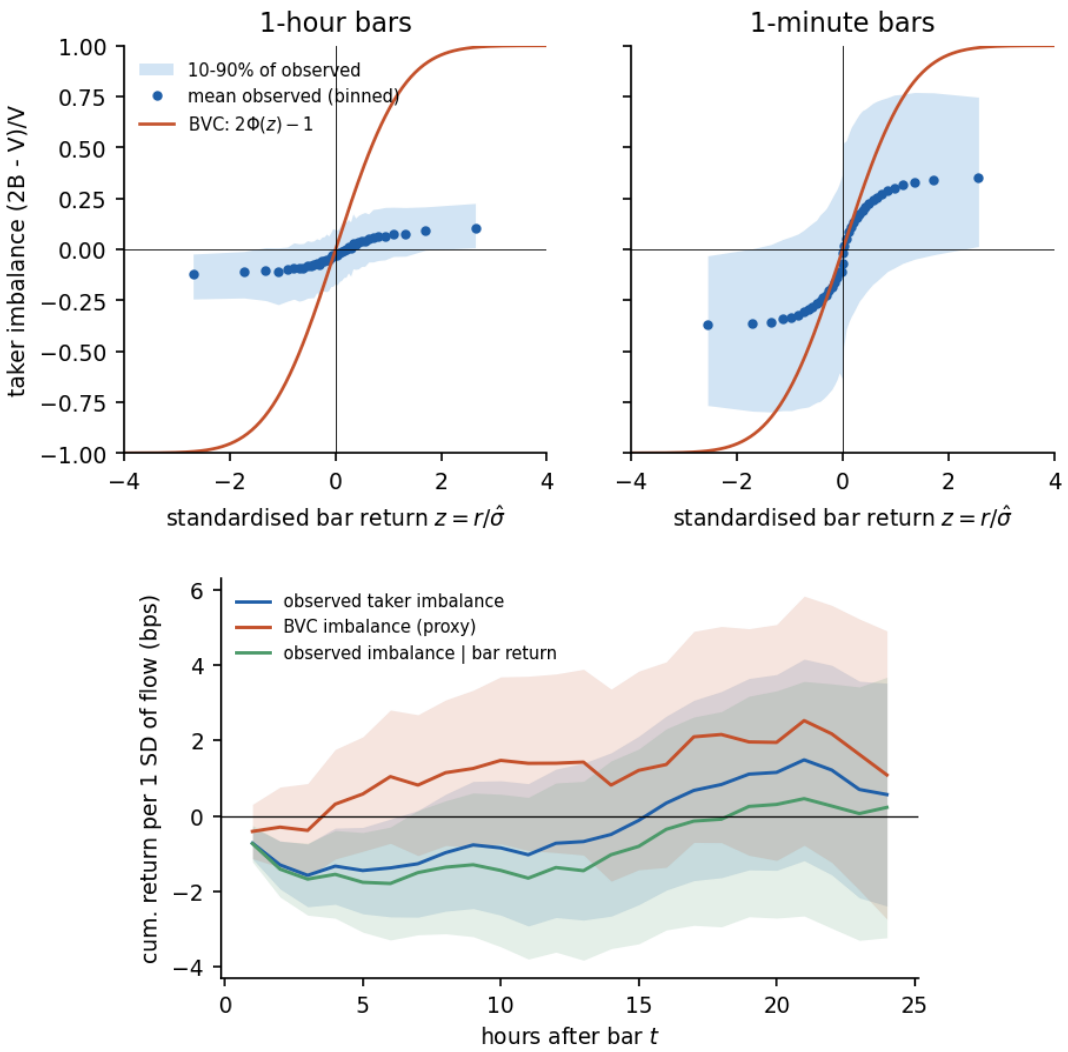


Figure 1: Top: observed taker imbalance against the standardised bar return (bin means and 10–90% band) with the BVC mapping $2\Phi(z) - 1$; BVC overstates the scale and cannot see most of the flow. Bottom: cumulative BTC return after a one-SD flow shock, for observed flow, hourly BVC flow, and observed flow holding the bar return fixed (95% Newey-West bands).

Why it matters. Bar-based flow and spread measures should not be read as measures of order flow or transaction costs in liquid electronic markets, and the predictive content of order flow at intraday horizons is precisely what such proxies discard. The natural next steps are order-book data at higher frequency, where flow imbalance at the best quotes is known to matter, and flow measured across venues.

Contact. hosseini.systems@gmail.com